

Graduate Research Plan Statement

During the Epoch of Reionization (EoR), the hydrogen in the Universe was (re)ionized by the first luminous sources – stars and accreting black holes in galaxies. The EoR marks the last major phase change of hydrogen in the Universe, and is intrinsically linked to the formation of the first stars and galaxies. Yet this era, critical to answering key cosmological questions such as how structure formed, is currently one of the most poorly understood times in cosmic history.

In the last decade, astronomers have begun to probe the EoR by characterizing galaxies that formed within a billion years after the Big Bang, though current observational facilities are limited in their ability (for example, by wavelength coverage, sensitivity, or atmospheric effects) to obtain large samples of these objects. However, the upcoming *James Webb Space Telescope* (*JWST*, planned to launch in late March 2021), will have the observational capabilities to detect galaxies that are **much fainter, less massive, and at much higher redshifts** than the current frontiers. *JWST* will also be able to **image in far greater detail than current facilities enable**. Together, these capabilities will extend the parameter space of galaxy evolution studies and enable astronomers to probe the properties of the earliest galaxies and the EoR in unprecedented detail.

Intellectual Merit:

I will use data from the *JWST* Advanced Deep Extragalactic Survey (JADES) to conduct the majority of my research in graduate school. JADES is a joint effort of the *JWST* Near Infrared Camera (NIRCam) and Near Infrared Spectrograph (NIRSpec) instrument teams, which will collaborate to take deep imaging and complementary spectroscopic data within the first few years of *JWST*'s operation. JADES has the potential to image thousands of galaxies at redshift $z > 6$ and tens of galaxies at redshift $z > 10$ (Williams et al., 2018), obtaining sample sizes an order of magnitude larger than current observations. Since JADES and other deep surveys, such as the Cosmic Evolution Early Release Science Survey (PI: Steven Finkelstein), will follow up imaging candidates with spectroscopic observations, we will be able to perform the **first analysis of the EoR and galaxy evolution during Reionization with large samples of spectroscopically confirmed galaxies**.

The Lyman alpha ($\text{Ly}\alpha$) spectral line of hydrogen is a sensitive probe of the ionization state of the Universe, as it has a large cross section for scattering by neutral hydrogen. Thus, the evolution of the flux distribution of $\text{Ly}\alpha$ from galaxies over cosmic time can be used as proxy for the quantity of neutral hydrogen in the Universe. However, current constraints on the EoR from the evolution of the $\text{Ly}\alpha$ emitter fraction are somewhat imprecise due to small sample sizes (e.g., Mason et al., 2018; Pentericci et al., 2018; Itoh et al., 2018; Jung et al., 2019). Thus, as a graduate student, I **propose to use the large quantity of high-quality spectroscopic data JADES will obtain to probe the EoR via observations of high-redshift galaxies**.

The JADES program is designed such that NIRCam imaging will be used to select sources for spectroscopic follow-up by NIRSpec. Hence, by applying color criteria to NIRCam observations, I will first identify candidate high-redshift galaxies that also have spectroscopic data obtained with NIRSpec. After reducing the data, ensuring the accuracy of our photometric redshift estimates ("low-redshift interlopers" must be identified and removed, as we cannot assume our sample is representative of the population of high-redshift galaxies if lower-redshift galaxies are present), and identifying galaxies

with reliable Ly α detections, I will characterize the observed Ly α lines (for example, the line shape and flux), which can be used to **probe properties such as star formation rate and neutral hydrogen covering fraction** of these high-redshift galaxies, which are currently poorly constrained. Furthermore, since neutral hydrogen attenuates Ly α , even non-detections of Ly α during Reionization can be used to probe the fraction of neutral hydrogen in the Universe. Therefore, I will also **construct an analysis framework to quantify the disappearance of Ly α as the Universe becomes increasingly neutral.**

This project is a natural extension my undergraduate research experience, and I have already developed some of the necessary tools for this work (for more details, see my personal statement). I have studied the EoR from multiple angles, both theoretical and observational, and am actively developing the technical and analytical skills necessary for data processing, analysis, and interpretation. Thus, this research plan allows me to **maximize my existing research experience to further the goal of understanding the early Universe.** As a graduate student working on Reionization-era galaxies, the University of Arizona is the ideal location for me to do this work. In particular, Professor Marcia Rieke (the PI of NIRC*am*) and Professor Dan Stark are JADES collaborators and have interests in galaxy evolution during the EoR.

Broader Impacts:

In graduate school, in addition to building on my undergraduate research, I will also continue my undergraduate outreach and mentoring activities. I will take advantage of organized outreach activities for the general public to **present my research to the public, showcasing scientific progress made possible by public investment in JWST.** If no opportunities for public outreach exist, I will work to create opportunities in the form of generally accessible talks, social events, or outreach events. However, I am particularly passionate about equity and inclusion in STEM, and as a graduate student, I **plan to be involved in a mentoring organization** akin to the Sundial Project at Arizona State University (described in my personal statement), and if no such organization exists, **I will start one.** By doing so, I hope to encourage underrepresented minority students (especially women) to be involved in research, and ultimately to make careers in physics and astronomy.

In graduate school, I plan to take full advantage of the great observational strides *JWST* will make, allowing astronomers to **probe the first stars and galaxies at a level of detail never seen before.** I will showcase my work at outreach events, **educating the general public** about the history of the Universe. I will also **develop support and mentoring systems aimed at undergraduate women** as they start their research careers, and by doing research at the frontier of extragalactic astronomy, will serve as a role model myself.

References

- Itoh R., et al., 2018, *The Astrophysical Journal*, 867, 46
Jung I., et al., 2019, *The Astrophysical Journal*, 877, 146
Mason C. A., Treu T., Dijkstra M., Mesinger A., Trenti M., Pentericci L., de Barros S., Vanzella E., 2018, *The Astrophysical Journal*, 856, 2
Pentericci L., et al., 2018, *Astronomy & Astrophysics*, 619, A147
Williams C. C., et al., 2018, *The Astrophysical Journal Supplement Series*, 236, 33